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Marine chronometers: the rapid adoption of new technology by East India captains in the period 1770–1792 on over 580 voyages

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Front cover: Dial of a small experimental floor-standing 8-day centre seconds regulator, which is discussed in Part 2 of Chris Watson's article on the horological connections of the John Smeaton family.
Photo courtesy of Bonhams, London.

Marine chronometers: the rapid adoption of new technology by East India captains in the period 1770–1792 on over 580 voyages

Simon C. Davidson*

This paper examines marine chronometer usage in the determination of longitude by East India Company captains. The original ships logs of 587 voyages by the East India Company for the period 1770–1792 were examined to determine the usage of marine chronometers by captains on voyages of eighteen to twenty-four months across three oceans to India and China. It was found that chronometers were adopted and routinely used by East India Company captains much earlier than has generally been supposed, being used on 80 per cent of all voyages at the end of the period. Chronometers displaced the use of lunars as the primary method of determining longitude sooner than has been claimed in recent assessments.

The start and end dates for this research were determined by:

- The first known use of a marine chronometer by Captain Charles Mears on the *HEICS Egmont* in voyage season 1769/70.¹ Mears was also one of the first two captains of the East India Company (EIC) maritime service to use the lunar distance method to determine longitude in 1763.²
- The introduction in 1791 by the EIC of a new form of pre-printed logbooks, to be used on the ships in its service, which had specific provision for longitude by chronometer to be entered daily.³ No direct instructions by the Committee of Shipping to captains of its ships that they must use chronometers can be found. These records were lost through neglect whilst in storage in the mid-nineteenth century. It is evident that by 1791 they were

expected to do so. In particular captains were required to present their logbooks and journals to the Committee of Shipping on the return of their voyage, together with charts of their ship's track showing 'longitude by observation and chronometer, which you are particularly directed to render as complete as possible'.⁴ It is not known when the Committee of Shipping promulgated this instruction but it is safe to assume it was soon after the publication of the pre-printed logbooks, especially when one considers the influence of the East India Hydrographer, as will be noted below. Also from 1787 the directors of the company were making a 'significant investment in accurate surveys of the Indian coast and its approaches, and showing considerable enthusiasm for, and faith in chronometers'.⁵

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1. Simon C. Davidson, 'The Use of Chronometers to Determine Longitude on East India Company Voyages', *The Mariner's Mirror* (2016), 102:3, 344–348.

2. Cambridge University Library, Board of Longitude Fair Minutes RGO 14/5, pp 83–5 and British Library (BL), India Office Records (IOR), L/MAR/B/535D *Egmont*, Capt. Charles Mears's Log, 1762/3.

3. W. E. May, 'The Log-books used by Ships of the East India Company', *Journal of Navigation*, Vol. 27 (1974), 116–118.

4. Charles Hardy, *A Register of Ships, employed in the service of the Honorable the United East India Company, from the year 1760 to 1810* (1813), Appendix Instructions to Commanders.

5. David Philip Miller, 'Longitude Networks on Land and Sea: The East India Company and Longitude Measurement 'in the Wild', 1770–1840', in *Navigational Enterprises in Europe and its Empires, 1730–1850*, ed. by Richard Dunn and Rebekah Higgett (London, 2016), pp. 223–247 (p. 239).



Fig. 1. An East India Company ship approaching Spithead in 1797 © British Library Board P2460.

John Harrison's ground-breaking innovation of an accurate marine timekeeper, that won for him an interim award in 1765 from the Board of Longitude after the sea trials of 1761–2 and 1763–4, started a technology race amongst precision watchmakers to complement and then supersede the lunar distance method of finding longitude, which had become available at the same time as Harrison's breakthrough in timekeeper design. Their purpose was to tap the large market amongst captains undertaking long voyages, who had started in 1763 to use the lunar distance method, further developed by Maskelyne and Waddington.⁶ This method had been proven but it was complex, involving long calculations

taking four hours,⁷ and was not available at all over the period of the new moon. It was only in the nineteenth century that significant improvements to calculation tables were made, simplifying the mathematics required for lunars.

The EIC did not provide chronometers nor charts to their ships, which from the late seventeenth century were all on charter from private owners over a number of voyages. Only a few were owned by the company at the beginning of the nineteenth century. Captains and officers had to provide their own charts and navigation instruments for their assigned voyages. Ships of the EIC undertook the longest voyages, crossing three oceans in each direction, voyaging to and from the East.

6. Nevil Maskelyne, *The British Mariners' Guide containing Complete and Easy Instructions for the Discovery of the Longitude at Sea and Land* (London, 1763) and Robert Waddington, *A Practical Method for finding the Longitude and Latitude of a Ship at Sea, by Observations of the Moon* (London, 1763)

7. Jane Wess, 'Navigation and Mathematics: A Match Made in the Heavens?', in *Navigational Enterprises in Europe and its Empires, 1730–1850*, ed. by Richard Dunn and Rebekah Higgitt (London, 2016), pp. 201–222. In her Conclusion, the author summarizes the situation: 'However the computation of lunar distances was tedious, complicated and opaque compared to calculations based on dead reckoning and without appropriate training remained beyond the capabilities of ordinary navigators'.

Voyage season	Number of voyages	Chronometers in use		Lunars in use		Dead reckoning only in use	
		Number	%	Number	%	Number	%
1769/70	30	1	3	8	27	21	70
1770/1	30	-----		3	10	27	90
1771/2	28	-----		6	21	22	79
1772/3	26	2	8	12	46	12	46
1773/4	16	-----		8	50	8	50
1774/5	16	1	6	2	13	13	81
1775/6	21	-----		11	52	10	48
1776/7	24	3	12.5	15	62.5	6	25
1777/8	21	-----		15	71	6	29
1778/9	20	2	10	13	65	5	25
1779/80	18	6	33	9	50	3	17
1780/1	21	-----		11	52	10	48
1781/2	26	3	12	18	69	5	19
1782/3	21	1	5	13	62	7	33
1783/4	16	4	25	9	56	3	19
1784/5	26	4	15	18	70	4	15
1785/6	43	9	21	31	72	3	7
1786/7	35	20	57	9	26	6	17
1787/8	34	17	50	14	41	3	9
1788/9	32	15	47	15	47	2	6
1789/90	31	20	65	9	29	2	6
1790/1	24	16	67	7	29	1	4
1791/2	28	23	82	5	18	-----	

Table 1. Analysis of methods used for longitude determination on EIC ships by voyage season.

It is necessary to set the background in the 1770s. Only John Arnold initially took up the challenge to produce a simpler and cheaper instrument to achieve the same result as John Harrison's H4. It is believed he started with a form of detached-cylinder escapement in his first movement and early in the decade developed the pivoted-detent escapement. However within ten years he was joined by a competitor in Thomas Earnshaw who was developing his spring-detent escapement. Thomas Earnshaw's escapement of 1781 was so successful that it continued to be used almost unaltered until chronometers were

replaced by the global positioning satellite system in the 1970s.

These early makers were able to produce marine chronometers for prices in the range £65–105, a sum that EIC captains could afford, as per voyage they were earning £5,000 to £15,000 through their permitted private trade.

Other makers joined Arnold and Earnshaw in the latter half of the 1780s, including Paul Philip Barraud, John Brockbank and Charles Haley.

Captain Charles Mears, one of the first two EIC captains to have used the lunar distance

method in 1763,⁸ decided he would give the new technology a trial in his voyage starting in 1770. We have no means of knowing if he was lent the chronometer in 1770 by Arnold or had purchased it. Quite probably the former was the case, as it was in Arnold's interest to pick a captain well versed in the lunar distance method, which was needed to check the rate of the chronometer. John Arnold would have known the Jerusalem Coffee House was the meeting place of EIC captains and ship owners. His contact with the Astronomer Royal, Nevil Maskelyne, would have alerted him to the name of Charles Mears, who gave evidence to the Board of Longitude in early 1765 of his use of this method which had been promoted in 1763 by Maskelyne and Waddington.⁹

The data in Table 1 show the voyage season, number of voyages of EIC ships, the number of ships using chronometers for the determination of longitude and their percentage of the total voyages that season. Also included for comparison are the numbers of ships using the lunar distance method and those solely using dead reckoning (see Appendix 1 for an explanation of lunar distance method and dead reckoning).

In all cases where the chronometer was in use on long passages between landings, lunars were taken periodically to check the rate of the chronometer, but this became much less frequent in the latter part of the 1780s, when captains' confidence in the going rate of chronometers appears to have increased with improvements the makers were making, especially as many captains owned more than one chronometer. It is important to note that there were EIC regulations about recording the ship's navigation and method used, so we can be confident that the logs are reasonably accurate in use. Although longitude determined by dead reckoning continued to be recorded in the logs even where chronometers or lunars were used, it is clear from the logs examined that the dead reckoning was corrected to reflect the longitude by either of the other methods. It was well recognized that dead reckoning on long passages between landfalls was prone to

many errors accumulated over those long passages, due to lack of knowledge of the currents.

In the 1770s, chronometers were not yet relied on for whole voyages. In this early period of testing, when the rate was checked by reference to known points of longitude (for example Tenerife / Cape of Good Hope / St Helena), or against results from lunars, chronometer rates were found to have changed. Sometimes they had ceased to run at all!

In the 1780s, when chronometers were proving more reliable in retaining a constant rate of gain or loss, they were much more commonly used throughout the voyage, and always when approaching known danger points such as coasts or changes of ocean current. In addition many captains were using several chronometers to determine longitude as it is more common to see the term chronometers (plural) where identifying chronometer numbers themselves are not given.

Throughout the period to 1792, where the lunar distance method was used to determine longitude with no chronometers present, it was only used approaching known danger points.

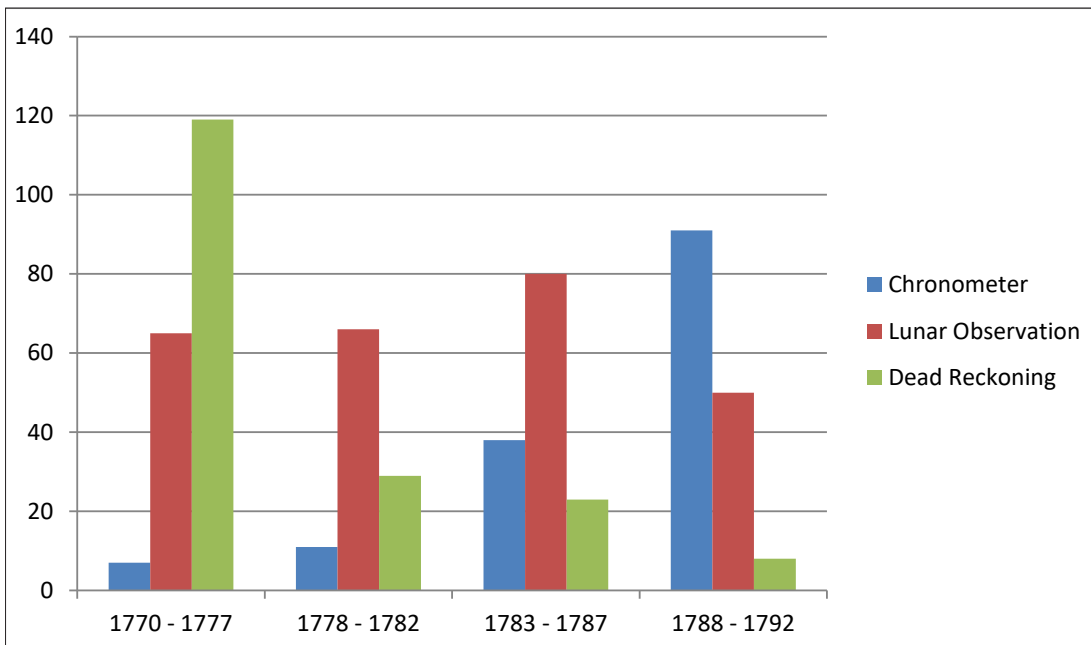
The danger points or changes of ocean current on these long voyages were:

- determining the turning point in the southern part of the north Atlantic, where there was a change in direction of the ocean current coupled with a position of little wind, in order not to be swept onto the Brazilian coast adding six to eight weeks to the voyage;
- approaching the Cape of Good Hope;
- the inside west passage, off Madagascar;
- the east passage off Madagascar when turning north for Bombay and Madras, or north east for China;
- entering the various straits through the Indonesian archipelago for China;
- approaching the Scilly Isles on return.

Even logs such as those of Joseph Huddart, who was well-known for his interest in

8. Davidson, 'The Use of Chronometers', p. 346.

9. Cambridge University Library, Board of Longitude Minutes.



Number of EIC ships using chronometers / lunars / dead reckoning to determine longitude.

navigation, show he only occasionally checked his longitude when using the lunar distance method in his voyage of 1778.¹⁰ However, once he had chronometers in his later voyages, he recorded longitude by them much more frequently and latterly almost daily.¹¹ The lunar distance method required several hours of mathematical calculations. It was a very tedious procedure and certainly rarely done daily.

Table 1 highlights that up to 1778/9 the following changes took place in the method used for the determination of longitude:

- dead reckoning was the sole method of determining longitude, 90 per cent of the time in 1770 falling to 25 per cent;
- use of the lunar distance method rose to 65 per cent;
- the use of chronometers, while not consistent in each voyage season, rose to 10 per cent of the time.

An analysis of the succeeding period from 1779/80 up to 1791/2 shows a significant rise in the use of chronometers to determine longitude:

- dead reckoning as the sole method continues to decline from 25 per cent to zero;
- the lunar distance method for determining longitude falls from 50 to 18 per cent;
- chronometers by 1786/7 were used on 50 per cent of all ships in the EIC maritime service and by the end of the period were used on 82 per cent of the ships to determine longitude.

The chart above illustrates the increasing rate of usage over four periods for the 587 voyages examined:

It is worthy of note that Alexander Dalrymple, Hydrographer to the East India Company (Fig. 2), published a pamphlet in various editions from 1780 to 1788: *Instructions concerning Arnold's Chronometers or Time-Keepers*.¹² Dalrymple's last pamphlet ran to twelve pages and even now reads very concisely, with detailed examples of how to use both box and pocket chronometers at sea.

- Dalrymple gives a detailed procedure of comparing the box chronometer with any

10. Captain's Log of the *Royal Admiral* BL, IOR: L/MAR/B/338A.

11. Captain's Log of the *Royal Admiral*, BL, IOR: L/MAR/B/338B, 338C & 338D.

pocket chronometers, to be used on deck when the altitude of the sun or star was to be taken. He specified two persons were needed: one to count the seconds on the box chronometer 'whilst the other sees the instant at which the small chronometer is at 60; which, by a little use, may be done by a glass, to the 10th of a second.'¹³

- He recommends a minimum of three, preferably five observations of the sun's (or star's) altitude using the time of the box chronometer.
- He explains how to calculate the longitude after adjusting apparent time to mean time, with examples.
- He urges checking the rate when in port for several days by taking altitudes of the sun and comparing the values to see if the rate of gain or loss of the chronometer has changed from that used in setting out on the voyage.
- He comments on the utility of checking the chronometer's rate on long passages at sea but cautions that making comparisons by lunar distances at sea was not precise enough to distinguish 'small errors' and only of use in correcting 'great errors' or at the end of a considerable period of time, when small daily errors may have become great by accumulation.
- He supplies 'A Form for Working the Observations of Time'.

Dalrymple's instructions on how to use chronometers continued to be published well after his death and right up to 1816.¹⁴

As Andrew Cook has demonstrated in his seminal paper on Alexander Dalrymple and John Arnold,¹⁵ the influence of the EIC's Hydrographer on the use of chronometers to make accurate charts and to locate position at

sea was highly significant in the 1780s. It also enabled navigators to take account of ocean currents particularly in the Atlantic where it was important to know when to change direction and head south-east in order not to be swept onto the coast of Brazil, with all the delays this entailed.

Dalrymple continued his advocacy of Arnold's chronometers in his general preface to a *General Collection of Nautical Publications* in 1783,¹⁶ which was then used to preface his later publications. In the quote below I have retained his use of capitals and italics.

The more recent Improvement in NAVIGATION, which leaves the *Lunar Observations* as far behind, as *they* did the *Variation* [he is referring to the variation of the isogonic magnetic lines running north to south which were used for a while to determine an approximate position], is by CHRONOMETERS; *Harrison* was the *first* who produced a TIME-KEEPER, really of use, to determine the *Longitude* at Sea; but the degree of precision to which *They* are now brought, by *Arnold*, would not, ten years ago, have been believed *within the bounds of possibility*...but since then Mr *Arnold* has made great Improvements, not the least of which seems to be the application of a *Gold Spring*.

The *only* Objection to *Chronometers* was the *Price*; those made by *Harrison*, or on his construction, were at a rate too high for common use, and although those made by *Arnold* were reduced to a fourth part of the Price, still 60 or 100 guineas made a sum effectually operating against the general use of those Machines: but as they are now sold by

12. Alexander Dalrymple, *Instructions Concerning Arnold's Chronometers or Time-Keepers* (London, May 1788). Various incomplete copies exist from 1780 but the most complete copies of the pamphlet of 1788 are in the Guildhall Library (GL), London: Clockmakers' Collection, 3.3 No.4 in 19; Institution of Civil Engineers Library, London: Vulliamy Collection Tract (4) 33; National Library of Scotland, Edinburgh: RB.m.357; University Library, Cambridge: Aaa.38. 12.

13. Dalrymple, *Instructions*, p. 2.

14. Andrew S. Cook, *Alexander Dalrymple (1737–1808), Hydrographer to the East India Company and the Admiralty, as Publisher*, PhD Thesis at University of St Andrews 1993, Vol. 2, Catalogue of Dalrymple's Letterpress Publications, p. 528.

15. Andrew S. Cook, 'Alexander Dalrymple and John Arnold: Chronometers and the Representation of Longitude on East India Company Charts', *Vistas in Astronomy*, Vol. 28, 1985, 189–195 (available at BL, IOR: ORW 1986.a.50).

16. BL: G.2197 (2), p 7-9.

Mr. *Arnold* at so low a price as 20 guineas, the general benefit will be even greater, than from his more perfect *Chronometers*, for in a short time few officers in the *Navy* or *India Service* will be without them, and the *mean* of *many such* will probably be as good a guide, as *one* more excellent, although for *Geographical Determinations* the *most* perfect will be desirable.

I think the CHRONOMETER of his Invention, deserves to have the name ARNOLD applied to it; and I shall use the name ARNOLD to imply such Machines. I do not presume to depreciate *Chronometers* on other constructions, but confine my opinion to the result of my own knowledge and experience.....Although I am a great Advocate in favour of CHRONOMETERS, I am far from depreciating *Lunar Observations*; on the contrary, I think the *Mean* of many *Lunar Observations*, connected by CHRONOMETER, might be usefully applied in determining the *rate* of the CHRONOMETER, in long periods at *Sea*, where there is no positive mode of determining *Its daily rate*.'

One could not envisage a greater advocacy for the use of chronometers and in particular those by Arnold than this by the Hydrographer to the East India Company. It is worth noting that the Admiralty did not appoint a Hydrographer until twelve years later and then they appointed Alexander Dalrymple in 1795. He then held both positions until his death in 1808—a unique situation.

John Arnold made the claim in the 1780s that there were rarely less than a hundred of his chronometers at sea.¹⁷ He was of course referring to the fact that voyages of two years overlapped and he was probably also supplying captains not in the East India maritime service. Appendix 2 lists the 147 voyages made where a chronometer was used to determine longitude with the name of the ship and its captain, with details of the number of chronometers, maker and number of the chronometer. In some cases a mate (1st M or 2nd M), the term used by the EIC for an officer, is shown either because it is only his log that exists today or in some cases

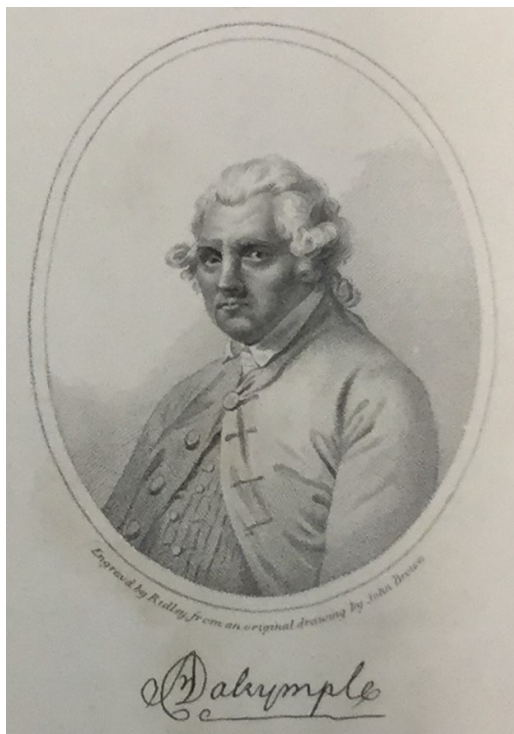


Fig. 2. Alexander Dalrymple (1737–1808), Hydrographer to the EIC 1779–1808 and Hydrographer to the Admiralty 1795–1808. © British Library Board P1470.

he submitted the log of his captain who perished on the voyage.

One noticeable change was that ships were successfully making the crucial turning point (from a direction of south-west to south-east) north of the equator to take advantage of the change in ocean currents and wind direction. This meant it became uncommon for EIC ships to touch the Brazilian coast, with consequently much shorter passages to India and China. If they missed the turning point, which happened frequently prior to the use of chronometers, it could add as much as two months to the voyage. Shorter passages were clearly more cost-efficient and the EIC, and particularly its Hydrographer, Dalrymple, were key drivers in such progressive navigational practice.

Analysing the names of the captains who used chronometers, one can discern from their backgrounds some possible reasons for a proportion of them to take up this method.

17. John Arnold, *Certificates and Circumstances Relative to the going of Mr Arnold's Chronometers* (London 1791) p iv, at GL: C.C. 3.3 No.3 in 19.

Wednesday the 12th day of May 1790
 chiefly a fresh trade with
 fair weather.
 At 4 Pm the Extremes of St. An-
 tonio @ S 8th E. to S. W. Dist^o 8th L^o
 and at 6 Pm it bore @ S 6th E to
 N. E. Dist^o 8th L^o Lagues
 Long. W. Box Chron. 1st 34. 25. 12
 2nd 36. 25. 26
 3rd 42. 25. 21
 allowing it to gain 3rd Day.
 Lat. N. 15. 3rd 1st

Fig. 3. Log of the William Pitt for 12 May 1790 listing box chronometers No. 54 and 86 and pocket chronometer No. 72 © British Library Board IOR/L/MAR/B/184C.

Thirteen of the captains who started to use a chronometer had previously been officers on ships where their captain was an early user of the lunar observation method, or they had served under a captain who was already using a chronometer.

Five of the captains using chronometers in the 1770s were members of the Society of East India Commanders (formed in 1773) which must have enabled the exchange of ideas, for example during the Society dinners between voyage seasons.

Unfortunately it is uncommon that the name of the maker is linked with the number of a chronometer—more often there is just mention of a box/large or small/pocket timekeeper/chronometer, and sometimes just a number. The term 'large' indicates a box chronometer, 'small' a pocket chronometer. An example of a description in a ship's log is shown in Fig. 3. The description in this case records the longitude per box chronometers No. 54 & No. 86 and also per pocket chronometer No. 72.

The term 'timekeeper' (sometimes 'timepiece' or 'watch') is more common in the earlier period of this study with the term 'chronometer' becoming more frequent in the 1780s. On some ships the term chronometer and timekeeper are interchanged in the same log for a given voyage. Until 1780 one can assume all the chronometers were made by John Arnold, and probably still the majority made by him through the 1780s.

The term 'timekeeper' was used following its use in earlier years by Harrison and others for the pioneering prototypes which led to the successful designs. The chronometer/timekeeper nomenclature question is covered by a footnote to a pamphlet Dalrymple published just before 24 March 1781, the day on which Joseph Banks received his baronetcy:

The Machine used for measuring Time at Sea is here named CHRONOMETER, my friend Mr. Banks agreeing with me in thinking so valuable a Machine deserves to be known by a Name, instead of a Definition. The name *Time-Keeper* is only proper to a *perfect Chronometer*.¹⁸

However an interesting comment by Captain William Fraser on the *Earl of Mansfield* in 1780, just after the first issue of Dalrymple's pamphlet, is worthy of note: 'My Timekeeper I shall call on this Voyage (agreeable to Mr Dalrymple's desire) the Chronometer'.¹⁹

Captain Robert Scott of the *Neptune* on 9 July 1780 records: 'Finding the Timepiece has lost upon a mean 6" per day since delivered to me by Mr Arnold both by observation of the sun and moon, Timepiece allowed accordingly'.²⁰

The use of a chronometer at this early date was not without disasters, an example being the comments of Captain Josiah Pryce in his log on the *Royal Charlotte* in 1784 en route from England to St Helena:

When going to adjust my Watch by the Chronometer this morning, to my great astonishment, found it at a stand. Conscious of my having wound it up yesterday prompted me to look at the works where to my loss I found the spring that supported the ballance [sic] broke and instrument rendered entirely useless.²¹

In 1786 Captain Robert Taylor of the *Earl of Talbot*, on 7 January off Table Bay, records in his log:

Found watches gained not half a second per day since 20 September instead of losing 1.5" as mentioned by Mr Arnold. Agreeable to Mr Arnold's directions I have allowed the chronometer to lose 1.5" per day which has made a small error in my longitude.²²

18. Alexander Dalrymple, *Some Notes useful to those who have Chronometers at Sea*, BL, G2198 (21); I am grateful to Dr Andrew Cook for drawing this to my attention.

19. BL: IOR: L/MAR/B/530B, Captain's Log, 7 June 1780.

20. BL: IOR: L/MAR/B/98F, Captain's Log, 9 July 1780.

21. BL: IOR: L/MAR/B/150-I, Captain's Log, 15 April 1784.

22. BL: IOR: L/MAR/B/410C, Captain's Log, 7 January 1786.

Taylor could be referring to a combination of pocket chronometers and/or a box chronometer, as he interchanges timekeeper/chronometer/watch throughout his log.

Captain Allen Chatfield of the *Rodney* records in his log some very positive results on 11 July 1787:

I find my Timepiece was set for the Meridian of Portsmouth make 16° W of Greenwich at noon. Instead of 17° 6' W. Portsmouth being 1° 6' W of Greenwich, discovered on arriving at Madeira.

Later in the voyage on arriving at Calcutta: 'On arrival at Calcutta I find my Timepiece set in making land only 14' to West'.²³

In his following voyage in the season 1789/90, Chatfield had similar excellent results from his chronometer on approaching St Helena after a voyage of over two months from Madras: 'Timepiece 1' West of St Helena'.²⁴

On the other hand, Captain James Dundas of the *Earl Fitzwilliam* on his return from Bengal to Madeira in 1791 was thoroughly disgusted with his chronometer: after comparing his longitude with that obtained by taking Lunar observations: 'Timekeeper good for nothing'.²⁵

This was in contrast to his previous voyage season 1786/7, when he had success navigating by determining longitude with a chronometer.²⁶

As mentioned at the beginning of this article, by the end of the 1780s Arnold and Earnshaw were joined by several other makers, and this must have had an influence on usage. Except for Earnshaw and Haley, they all had their premises close to the haunt of the East India trade at the Jerusalem Coffee House in Cowper Court, to the south side of Cornhill near the Royal Exchange.

Long-term survival

In the 147 voyages of ships using chronometers, Arnold's chronometers were referred to in 33 voyages and there is a voyage mentioning a Barraud chronometer. Specific numbered chronometers were referred to in only 14 voyages, although where they were mentioned several numbered chronometers were often listed (see Appendix 2).

The unique numbering of chronometers offers some clues for tracking their survivorship over the long term. Reviewing all the numbered Arnold chronometers, and additionally assuming that all the numbered but not named chronometers were made by Arnold, Hans Staeger's listing reveals the whereabouts of a number of relevant chronometers:²⁷

- Arnold box chronometers nos 11, 16 and 86 appear to exist in private collections; no. 1 in the Clockmakers' Museum, London; no. 12 in the British Museum, London, and no. 95 in the Marine Museum, Washington D.C.
- Arnold pocket chronometers nos 25, 72 and 354 appear to exist in private collections; no. 11 in the British Museum, London, and no. 518 in the Science Museum, London.

Conclusions

It is apparent that during the latter decades of the eighteenth century, the use of new technology—the marine chronometer as the primary method to determine longitude—displaced the lunar observation method that had been introduced into the EIC maritime service in 1763. In the space of twenty-two years from 1770 the marine chronometer became the primary method.

Given the simplicity and efficacy in the use of the chronometer, one might ask why it took so long to make the change in the method of

23. BL: IOR: L/MAR/B/442E Captain's Log, 11 March 1787 and 29 June 1787.

24. BL: IOR: L/MAR/B/442F Captain's Log, 6 April 1791.

25. BL: IOR: L/MAR/B/ 491B, Captain's Log, 5 January 1791.

26. BL: IOR: L/MAR/B/491A Captain's Log, 1787.

27. Hans Staeger, *100 Years of Precision Timekeepers from John Arnold to Arnold and Frodsham 1763–1862* (Stuttgart, 1997).

determining longitude? The answer lies in the length of the voyages at that time—eighteen to twenty-four months—and then the success of the method had to be conveyed largely by word of mouth to colleagues who might not make their next voyage for another one or two years. Importantly, a significant actor such as Captain Charles Mears, who was a pioneer in the use of the lunar distance method for determination of longitude, was then also a pioneer in the use of marine chronometers.

One could also ask why was it that the maritime service of the EIC was so far ahead of Royal Navy and the wider maritime world in taking up the new navigational technology of chronometers? I would suggest that this is answered by a combination of factors:

- The maritime service of the EIC, unlike other shipping, were transporting very high value cargoes equivalent in today's money to £10-20 million per ship over three oceans with many hazards and uncharted reefs. This was a powerful stimulus to drive new technologies in navigation such as the adoption of chronometers to determine longitude, offering a much easier and faster method.
- There had been great interest from the EIC in the 1760s in the, then new, lunar distance method of determining longitude influenced by Nevil Maskelyne, who had strong connections to the EIC through his wife, the sister-in-law of Lord Clive, a prominent figure in the EIC in the 1760s. In the 1770s and later, Maskelyne as Astronomer Royal and a member of the Board of Longitude, was well placed to promote the use of chronometers for the determination of longitude by the EIC maritime service.
- The maritime service of the EIC was a close-knit community that met between voyages at the Jerusalem Coffee House in Cowper Court, where the Society of East India Commanders held their meetings. It was also where they met the ship owners and recruited their junior officers. Significantly, all but two chronometer makers established their workshops in this area.
- Alexander Dalrymple, Hydrographer to the East India Company, who had significant influence on both the EIC directors and the maritime service, was from 1780 an active advocate of the use of chronometers to determine longitude. There was no such figure or position in the Royal Navy until the Admiralty subsequently appointed Dalrymple as Hydrographer to the Admiralty in 1795.
- The captains of the EIC had sufficient income to purchase chronometers to safeguard their cargoes and their own private trade goods aboard the ship.
- Captains in the Royal Navy did not have the same income and did not have the responsibility for transporting high value cargoes across three dangerous oceans with uncharted hazards and currents. Until Dalrymple's appointment in 1795, they also had no influential person in the Admiralty encouraging the use of chronometers on all voyages. The Admiralty only provided chronometers for survey and for exceptionally long voyages, such as those by Cook and Vancouver, and did not consider their utility for general use until after 1820.

Chronometers were adopted and routinely used somewhat earlier by EIC captains than has been supposed, and became the primary method over the use of lunars sooner than has been claimed in recent assessments.²⁸ By 1792 marine chronometers were in use for the determination of longitude on just over 80 per cent of annual voyages by EIC ships. I examined a sample of logs over the following years from 1792 and found the 80 per cent usage of chronometers was maintained and in some years exceeded. Since at this period the EIC had over one hundred ships in service at any one time, one realises that at a minimum there were eighty chronometers in use. However, by this time, captains who were using chronometers generally had three, sometimes four, which gives a total of up to 240 chronometers in use, owned by captains in the maritime service of the EIC.

28. Richard Dunn & Rebekah Higgitt, *Ships, Clocks and Stars: The Quest for Longitude* (2014) and Wess, 'Navigation and Mathematics' (see note 7).

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complete paper. I would also like to thank the British Library India Office Records staff responsible for document retrieval for supplying me with over 900 heavy ship's logs over a period of eighteen months.

Appendix 1

The lunar distance method (lunars)

This method used the moon's apparent motion relative to the sun or stars like a clock to find a reference time, e.g. the time at Greenwich. The observer used an octant, later the sextant as it became more widely available, to measure the angle between the moon and star or sun and their elevations. The method involved considerable calculation following the observation to make allowance for atmospheric refraction, owing to bending of light rays in the atmosphere and for lunar parallax, because the observations are made from the earth's surface not its centre, whereas the tables of the position of the moon refer to the centre of the earth. The observer then consulted prepared tables of lunar distances and the times at which they would occur. The difference between this time and local time on board the ship, determined from the altitude of the sun or star (also needed for determination of longitude by chronometer), could be converted into a difference of longitude, one hour of time difference equating to 15° of longitude. Nevil Maskelyne and his assistant, Robert Waddington, had both tried out tables developed by Tobias Mayer on a voyage they made in 1761 to and from St Helena and separately published tables of lunar distances on their return in 1763.

Once the observations had been made, it then involved making mathematical calculations which would take several hours to achieve a result. One severe limitation was that this method could not be used around the period of a new moon.

Dead reckoning

This method involves computing from the distance travelled, using the speed of the ship measured by the log line, keeping a record of the ship's heading (using the compass), assessment of currents and winds to determine the distance travelled from the last known departure from land. It was prone to errors, particularly in relation to the effects of currents which were poorly understood until the mid-nineteenth century. Cautious captains used to 'heave to' overnight when approaching what they thought were coastlines.

Appendix 2

Listing the 147 voyages made in the period 1769/70 and 1791/2 where a chronometer was used to determine longitude, with the name of the ship and its captain, with details of the number of chronometers, maker and the identifying number of the chronometer. '1st M' or '2nd M' stands for first or second mate, the term used by the EIC for an officer.

Main reference is from ship's logs at British Library India Office Records, L/MAR/B/ : one to three numbers followed by a letter.

- References with letter /A refer to page in: J. Arnold, *Certificates and Circumstances Relative to the going of Mr Arnold's Chronometers* (London, 1791) with page number. Also one mention of an Arnold box chronometer belonging to Captain James Dundas of the Prime 1778/9 on p. 14 of *An Answer from Mr Arnold to An Anonymous Letter* (London, 1782), GL: C.C 3.3 No.2 in 19.
- References with letter /D refer to page in: A. Dalrymple, *A Full Answer to the Advertisement concerning Mr Earnshaw's Timekeepers* (London, 1806), BL: Gen. Ref. Coll. B.519 (12)
- Reference with letters DM followed by page numbers refers to: A. Dalrymple, *Memoir of a Chart of the Passage to the Eastward of Banka by Captain Lestock Wilson 1789*, BL: G.2197(2). I am indebted to Dr Andrew Cook for this reference.
- Reference /M refers to page in: V. Mercer, *John Arnold & Son*(1972)

Voyage season	Ship	Captain	Instrument used	References
1769/70	<i>Egmont (2)</i>	Charles Mears	Arnold	535F
1772/3	<i>Bridgewater (2)</i>	Nicholas Skottowe	Chronometer	42F
"	<i>Egmont (2)</i>	Charles Mears	Arnold	535G/A2-3
1774/5	<i>Grenville</i>	Burnet Abercrombie	Arnold	467D/D69
1776/7	<i>Egmont (2)</i>	Charles Mears	Arnold	535H
"	<i>Valentine (2)</i>	James Ogilvie	Chronometer	452F
"	<i>Princess Royal (2)</i>	Robert Kerr	Arnold	405F
1778/9	<i>Prime</i>	James Dundas	Arnold Box	509D-E/A7
"	<i>Princess Royal (2)</i>	Robert Kerr	Chronometer	405G-H
1779/80	<i>Bridgewater (2)</i>	William Parker	Chronometer	42H
"	<i>Earl of Mansfield</i>	William Fraser	Chronometer	530B
"	<i>Glatton (2)</i>	John Clements	Chronometer	172G
"	<i>Royal Admiral (1)</i>	Joseph Huddart	Arnold	338B
"	<i>Neptune (4)</i>	Robert Scott	Arnold	98F-G
"	<i>Bellmont</i>	William D Gamage	Chronometer	444A
1781/2	<i>Britannia (4)</i>	Edward Cumming	Chronometer	285MM
"	<i>Antelope (4)</i>	Henry Wilson	Chronometer	570A
"	<i>Brilliant</i>	Charles Mears	Arnold	no log/A39
1782/3	<i>General Eliott</i>	Robert Drummond	Arnold	269A/A31
1783/4	<i>Royal Admiral (1)</i>	Joseph Huddart	Arnold	338C/D63
"	<i>Earl Cornwallis</i>	Thomas Hodgson 1st M	Chronometer	348A
"	<i>Royal Charlotte (2)</i>	Josiah Pryce	Arnold No.1	150-I
"	<i>Earl of Mansfield</i>	William Fraser	Chronometer	530C
1784/5	<i>Britannia(4)</i>	Edward Cumming	Arnold No. 80	285NN/A32
"	<i>Dutton (2)</i>	James West	Chronometers	505F
			No. 75,83,95	
"	<i>Earl of Talbot (1)</i>	Robert Taylor	4 x Arnold	410C
"	<i>Dublin</i>	William Smith	Chronometer	399A
1785/6	<i>Earl Cornwallis</i>	Thomas Hodgson	Chronometer	348B
"	<i>Bridgewater (3)</i>	William Parker	Chronometer	42-I
"	<i>Carnatic (1)</i>	Lestock Wilson	Chronometer	165A
"	<i>Pitt (2)</i>	George Cowper	Chronometer	525D
"	<i>Royal Charlotte (2)</i>	Josiah Pryce	Chronometer	150J
"	<i>Stormont (2)</i>	George Curtis	Arnold	458G
"	<i>Manship (1)</i>	Charles Gregoire	Chronometer	363A
"	<i>William Pitt (1)</i>	Charles Mitchell	Chronometer	184A
"	<i>Kent (5)</i>	Richard Hardinge	2 x Chronometers	41C
1786/7	<i>Britannia (4)</i>	Edward Cumming	Chronometer	285OO
"	<i>Earl Fitzwilliam</i>	James Dundas	Chronometer	491A
"	<i>Princess Royal (3)</i>	James Horncastle	Chronometer	405-I
"	<i>Atlas (1)</i>	Allen Cooper	Arnold Pocket	27D
"	<i>Rodney (2)</i>	Allen Chatfield	Chronometer	442E
"	<i>Thetis (1)</i>	Justinian Nutt	2 x Arnold	387A/M104
"	<i>General Eliott</i>	Robert Drummond	Chronometer	269C
"	<i>King George (4)</i>	George Millett	Chronometer	402F
"	<i>Melville Castle</i>	Philip Dundas	Chronometer	268A
"	<i>Admiral Barrington</i>	Charles Lindgreen	Chronometer	512A
"	<i>Belvedere</i>	William Greer	Chronometer	332A
"	<i>Earl of Wycombe</i>	John W Wood	Chronometer	409A
"	<i>Lord Walsingham</i>	John Paiba	Chronometer	346A

Voyage season	Ship	Captain	Instrument used	References
1786/7	<i>Nottingham (4)</i>	Archibald Anderson	Chronometer	287G
"	<i>Busbridge</i>	Thomas Robertson	Chronometer	413B
"	<i>Glatton(2)</i>	Charles Drummond	Arnold	172J
"	<i>Houghton (5)</i>	James Monro	Chronometer	438Q
"	<i>Osterley (3)</i>	Joseph Clarkson	Chronometer	400K
"	<i>Rose</i>	John H Dempster	Chronometers	59A
			Box No.28, Pocket gold 85	
"	<i>Royal Admiral (1)</i>	Joseph Huddart	Arnold No.11 & 16	338D/A41
1787/8	<i>Barwell</i>	Thomas Welladvice	Arnold Box	420C/A43
"	<i>General Coote</i>	James Baldwin	2 x Arnold	441C/A40
"	<i>Earl Cornwallis</i>	Thomas Hodgson	Chronometer	348C
"	<i>Asia (3)</i>	John D Foulkes	Chronometer	24G
"	<i>Manship (1)</i>	Charles Gregoire	Arnold	363B/A43
"	<i>Duke of Montrose</i>	Joseph Dorin	2 x Chronometer	775C
"	<i>Dublin</i>	William Smith	Chronometer	399B
"	<i>Winterton</i>	George Dundas	Chronometer	451C
"	<i>Swallow (3)</i>	Robert Anderson	Chronometer	385E
"	<i>William Pitt (1)</i>	Charles Mitchell	Box & Pocket	184B
"	<i>Phoenix (3)</i>	Alexander Gray	Chronometer	175B
"	<i>Dutton (2)</i>	Thomas Barnard 2nd M	Chronometer	505G
"	<i>Deptford (2)</i>	John Gerrard	Chronometer	426G
"	<i>Earl Talbot (1)</i>	John Woolmore	Arnold	410D
"	<i>Boddam</i>	Joseph Elliott	Chronometer	351A
"	<i>Bridgewater (3)</i>	William Parker	Arnold Box. Pocket x 2	42J
"	<i>Lord Hawkesbury</i>	John Barkley	>1 Chronometer	323A
1788/9	<i>Vansittart (3)</i>	Lestock Wilson	Arnold Box No.56	46H/DM17-8
			Barraud Box No. 48	DM17-8
			Arnold Pocket No.8/309	DM17-8
			Arnold Box owned by EIC	DM17-8
		Walter Carruthers 1st M	Arnold Pocket 141	DM17-8
"	<i>Busbridge</i>	Thomas Robertson	Chronometer	413C
"	<i>General Goddard</i>	Thomas Foxall	Chronometer	440D
"	<i>Rockingham (1)</i>	John A Blanshard	Chronometer	124B
"	<i>Thetis (1)</i>	Justinian Nutt	Arnold Box No.24	387B/DM33
"	<i>Earl of Wycombe</i>	John William Wood	Chronometer	409B
"	<i>King George (4)</i>	John Sherwood	Chronometer	402H
"	<i>Melville Castle</i>	Philip Dundas	Chronometer	268B
"	<i>Rose</i>	John H Dempster	Arnold Box	59B
"	<i>General Elliott</i>	Robert Drummond	Chronometer	269D
"	<i>Lascelles</i>	Richard A Farington	Chronometer	408D
"	<i>Earl of Chesterfield</i>	Henry Burges	Chronometer	507J
"	<i>Valentine (3)</i>	John Lewis	Box & Pocket	452J
"	<i>Pitt (2)</i>	Edward Manning	Chronometer	525E
"	<i>Britannia (4)</i>	Edward Cumming	Chronometer	285PP
1789/90	<i>Earl Fitzwilliam</i>	James Dundas	Chronometer	491B
"	<i>Swallow (3)</i>	George Curtis	Chronometer	385F
"	<i>Hawke (5)</i>	Richard Pennell	Chronometer	390K

Voyage season	Ship	Captain	Instrument used	References
1789/90	<i>William Pitt (1)</i>	Charles Mitchell	Chronometers	184C
			Box No.54, 86, Pocket No.72	
"	<i>Rodney (2)</i>	Allen Chatfield	Chronometer	442F
"	<i>Royal Charlotte (3)</i>	Josiah Pryce	Chronometer	150K
"	<i>Belvedere</i>	William Greer	Chronometer	332B
"	<i>Princess Amelia (3)</i>	George Millett	Chronometer	36H
"	<i>Lord Walsingham</i>	James Young	Chronometer	346B
"	<i>Warren Hastings (1)</i>	John P Larkins	2 x Chronometer	9E
"	<i>Princess Royal (3)</i>	James Horncastle	Chronometer	405J
"	<i>Worcester (4)</i>	John Hall	Arnold Box & Pocket	278K
"	<i>Barwell (1)</i>	Thomas Welladvice	Arnold Box	420D
"	<i>Francis (2)</i>	Robert Burrowes	Chronometer	443D
"	<i>Earl Talbot (1)</i>	John Woolmore	Chronometer	410E
"	<i>Royal Admiral (1)</i>	Essex H Bond	Chronometer	338E
"	<i>Ceres (2)</i>	George Stevens	Chronometer	215G
"	<i>Hindostan (1)</i>	William MacKintosh	Chronometer	267A
"	<i>Sir Edward Hughes</i>	Robert Anderson	Arnold	354A
"	<i>Triton (3)</i>	William Agnew	2 x Chronometer	366J
1790/1	<i>Manship (1)</i>	John Lloyd	Chronometer	363C
"	<i>Dublin</i>	William Smith	Chronometer	399C
"	<i>Carnatic (2)</i>	John Corner	Chronometer	165C
"	<i>Phoenix (3)</i>	Alexander Gray	Chronometer	175C
"	<i>Kent (5)</i>	Richard Hardinge	Chronometers	41E
"	<i>Dutton (2)</i>	James Hamilton	Chronometers	505-I-J
		Thomas Barnard 2nd M	No.12,25,237,354	
"	<i>Britannia (4)</i>	Edward Cumming	Chronometer	285QQ
"	<i>Taunton Castle</i>	James Urmiston	Chronometer	107A
"	<i>Bridgewater (3)</i>	Robert Reay 1st M	Chronometer	42K
"	<i>Boddam</i>	John Jones	Chronometer	351B
"	<i>Earl Cornwallis</i>	Thomas Hodgson	Chronometers	348D
			No.7,10,261	
"	<i>General Coote</i>	James Baldwin	Chronometer	441D
"	<i>True Briton (4)</i>	Henry Farrer	Chronometer	297L
"	<i>Lord Hawkesbury</i>	John Barkly	Box & Pocket	323B
"	<i>Alfred (2)</i>	James Farquharson	Chronometer	140F
1791/2	<i>Duke of Montrose</i>	Joseph Dorin	Chronometer	775C
"	<i>Rockingham (1)</i>	John A Blanshard	Chronometer	124C
"	<i>Thetis (1)</i>	Justinian Nutt	Chronometers	387C
			No.24, 199	
"	<i>Earl of Wycombe</i>	John W Wood	Chronometer	409C
"	<i>Melville Castle</i>	Philip Dundas	Chronometer	268C
"	<i>Rose</i>	John H Dempster	Chronometer	59C
"	<i>Bellmont</i>	James Stewart 1st M	Chronometer	444G
"	<i>Earl Talbot (1)</i>	Jeremiah Dawkins	Chronometer	410F
"	<i>General Goddard</i>	Thomas Wakefield	Chronometer	440E
"	<i>Duke of Buccleugh</i>	Thomas Wall	2 x Chronometers	132B
"	<i>Middlesex (2)</i>	John Rogers	Chronometer No.518	450H
"	<i>Sullivan</i>	Robert Pouncy	Chronometer	257D
"	<i>Lascelles</i>	Richard A Farington	Chronometer	408E
"	<i>Busbridge</i>	Thomas Robertson	Arnold	413D
"	<i>Ganges (1)</i>	Joseph Garnault	Chronometer	86E
"	<i>King George (4)</i>	Richard Colnett	Chronometer	402-I

Voyage season	Ship	Captain	Instrument used	References
1791/2	<i>Europa (2)</i>	Augustus Applegarth	Chronometer	425H
"	<i>Lord Macartney</i>	James Hay	Chronometer	415D
"	<i>Nottingham (4)</i>	John Barfoot	Chronometer	287J
"	<i>Sir Edward Hughes</i>	Robert Anderson	Chronometer	354B
			& EIC Arnold No.44	
"	<i>Royal Admiral (1)</i>	Essex H Bond	Chronometer	338F
"	<i>Valentine (3)</i>	Iver McMillan	Chronometer	452K
"	<i>Ocean (1)</i>	Andrew Patton	Chronometer	222B